A 1.3.6-7 Control Systems (HL Only)



A 1.3.6-1.2.7 Control Systems

Control systems are automated decision-makers with controllers ("brains"), sensors (gather info), actuators (take action), and algorithms (rules). They use input/output/feedback to achieve goals. Open-loop systems lack feedback, while closed-loop systems use it for accuracy. Examples include self-driving cars, thermostats, and elevators.

A1.3.6 Components and Mechanisms

Imagine a self-driving car navigating a busy street. It needs to perceive its surroundings, make decisions, and control its movements. This is achieved through a control system with these key components:

Controllers: The "brain" of the system, making decisions based on input and desired output (e.g., the car's onboard computer).
Sensors: Gather information about the system's environment (e.g., cameras, lidar, radar).
Actuators: Carry out the controller's commands to affect the system (e.g., steering, throttle, brakes).
Transducers: Convert one form of energy to another (e.g., a camera converting light into an electrical signal).
Control Algorithms: The set of rules or calculations the controller uses to make decisions (e.g., the car's navigation algorithms).

Key Question: Can you think of another real-world example of a control system and identify its components?

Input, Process, Output, and Feedback

Control systems operate through a continuous cycle:

- **Input**: The desired value or setpoint (e.g., the desired speed for the car).
- **Process**: The system being controlled (e.g., the car itself).
- **Output**: The actual value of the controlled variable (e.g., the car's current speed).
- **Feedback**: Information about the output is fed back to the controller, allowing it to adjust its actions (e.g., the car's speed sensor provides feedback to the controller).

Open-Loop vs. Closed-Loop Systems

Open-loop: The controller sends commands without considering the actual output (e.g., a timer-based sprinkler system).
Closed-loop: The controller uses feedback to adjust its commands based on the actual output (e.g., the car's cruise control).

Key Question: What are the advantages and disadvantages of open-loop and closed-loop systems?

A1.3.7 Real-World Applications

Control systems are pervasive in modern technology, and found in various applications:

Autonomous Vehicles: Sensors gather data about the environment, and the controller uses algorithms to navigate and control the vehicle.
Home Thermostats: Maintain a desired temperature by sensing the current temperature and adjusting the heating/cooling system.
Automatic Elevator Controllers: Efficiently transport passengers between floors using sensors to determine position and weight.
Automatic Washing Machines: Control the washing cycle based on sensor input about water level, temperature, and load size.
Traffic Signal Control Systems: Regulate traffic flow using sensors to detect vehicle presence and optimize light timing.
Irrigation Control Systems: Efficiently water crops or lawns based on sensor input about soil moisture and rainfall.
Home Security Systems: Detect and deter intruders using sensors to trigger alarms, lights, or cameras.
Automatic Doors: Open and close automatically based on sensor input, such as motion detection.

Quiz Control Systems

1

Which component of a control system is responsible for gathering information about the environment?

A. ○ Sensor

B. ○ Algorithm

C. ○ Actuator

D. ○ Controller

2

What is the role of a transducer in a control system?

A. ○ To convert energy

B. ○ To make decisions

C. ○ To gather information

D. ○ To perform actions

3

Which type of control system uses feedback to adjust its actions?

A. ○ Straight-feed

B. ○Feed-loop

C. ○ Closed-loop

D. ○ Open-loop

4

Which of the following is NOT an example of a control system?

A. ○ A traffic light

B. ○ An automatic washing machine

C. ○ A home thermostat

D. ○ A manual door

5

What is the "input" in a control system?

A. ○ The action taken by the system

B. ○ The feedback from the system

C. ○ The desired state of the system

D. ○ The current state of the system

6

Which of the following is an advantage of a closed-loop system?

A. ○ Accuracy

B. ○ No need for sensors

C. ○ Low cost

D. ○ Simplicity

7

What is the role of the controller in a control system?

A. ○ To convert energy

B. ○ To make decisions

C. ○ To perform actions

D. ○ To gather information

8

Which of the following is an example of an actuator in a car's cruise control system?

A. ○ Steering wheel

B. ○ Throttle

C. ○ GPS

D. ○ Speed sensor

9

Which control system is more likely to be used in a safety-critical application?

A. ○ Both are equally suitable

B. ○ Neither is suitable

C. ○ Closed-loop

D. ○ Open-loop

10

Which of the following is NOT a real-world application of control systems?

A. ○ Autonomous vehicles

B. ○ Automatic doors

C. ○ Home security systems

D. ○ Manual can opener